Prajyot Shinde D20A Roll no.57

Blockchain Experiment 06

Aim: Creating Smart Contracts and performing transactions using Solidity and Remix IDE

Theory:

A Smart Contract is a self-executing program stored on the blockchain that automatically enforces the terms of an agreement when predefined conditions are met. It eliminates the need for intermediaries, ensuring transparency, security, and trust between parties. Smart contracts are written in Solidity (for Ethereum-based systems) and executed on the Ethereum Virtual Machine (EVM).

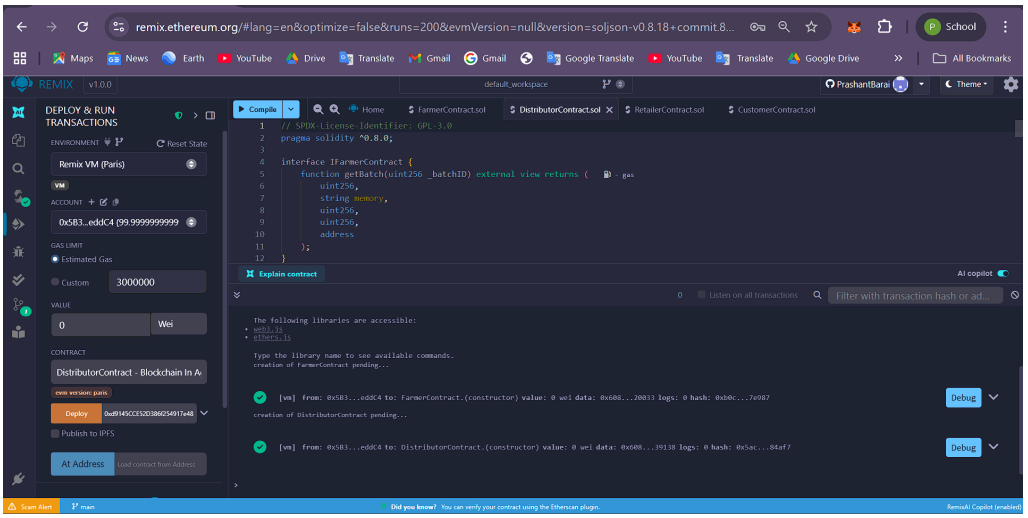
Significance of Smart Contracts in the Ethereum Blockchain:

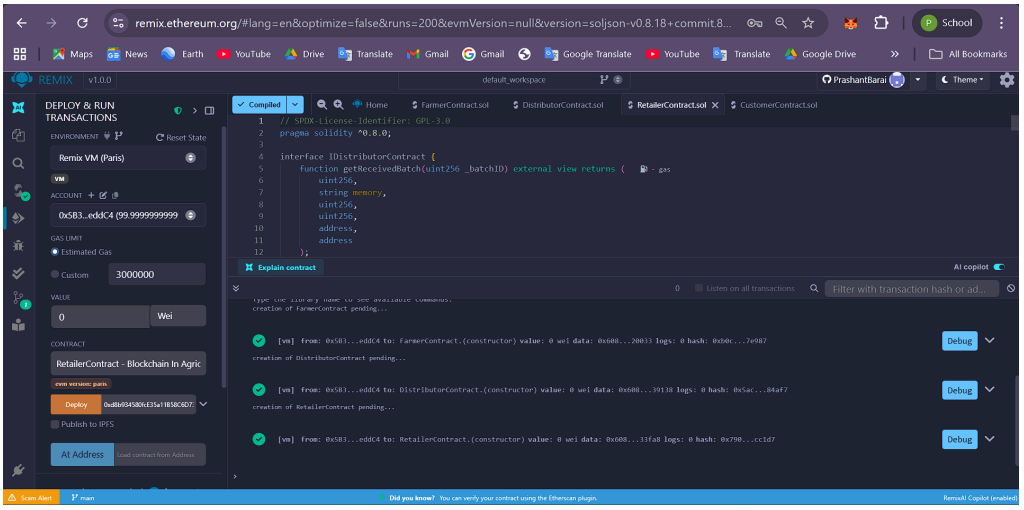
- Automation: Executes actions automatically when conditions are satisfied.
- Transparency: The contract code and transactions are visible on the blockchain.
- Security: Once deployed, contracts are immutable and tamper-proof.
- Cost-effective: Removes intermediaries and reduces transaction costs.
- Trustless System: Participants can interact without needing to trust each other directly.

Smart contracts form the backbone of decentralized applications (DApps) and decentralized finance (DeFi) systems on Ethereum.

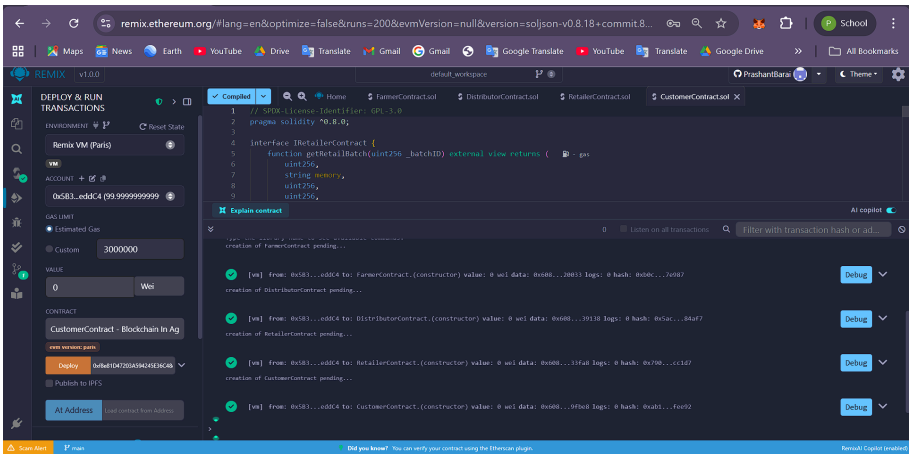
Implementation:

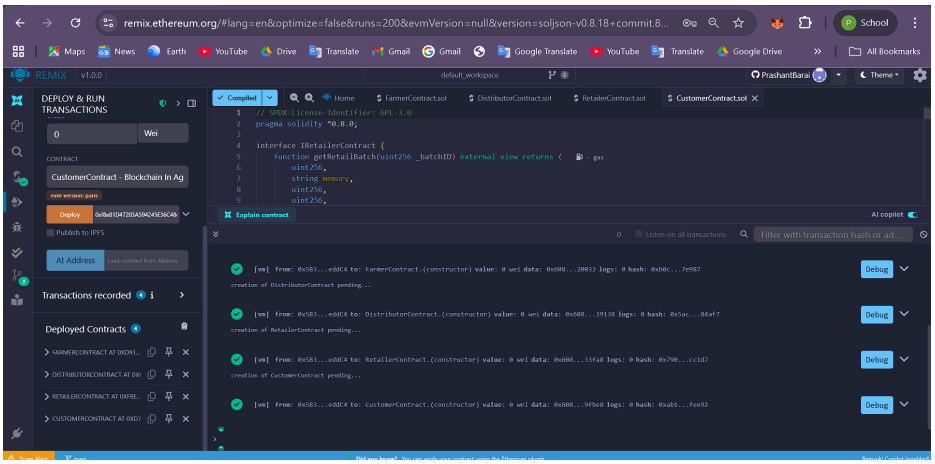
Output:

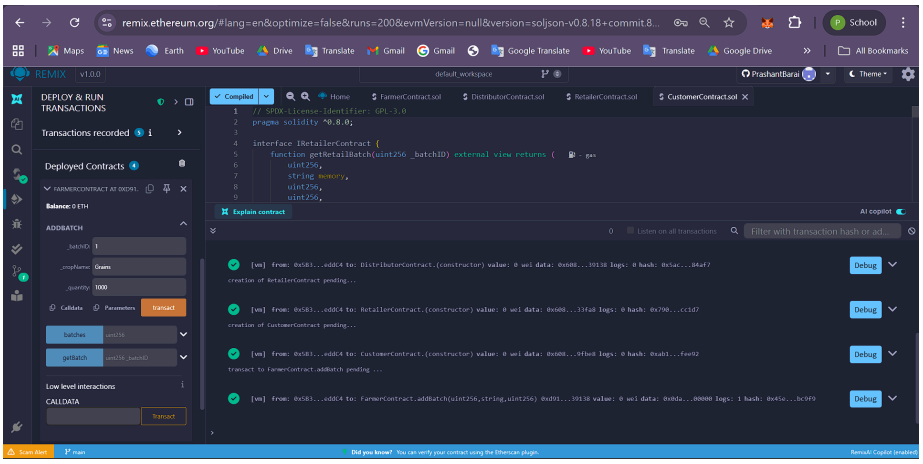


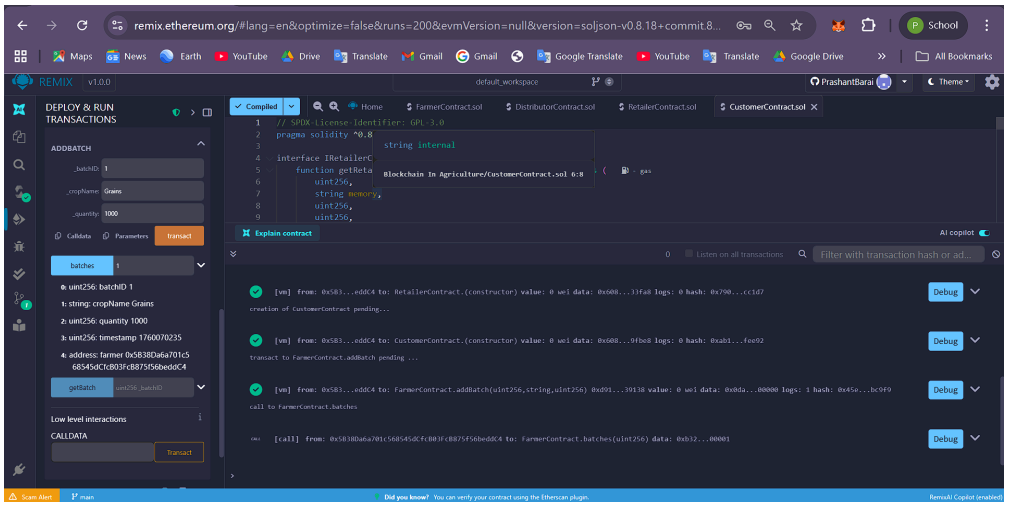


Prajyot Shinde D20A Roll no.57

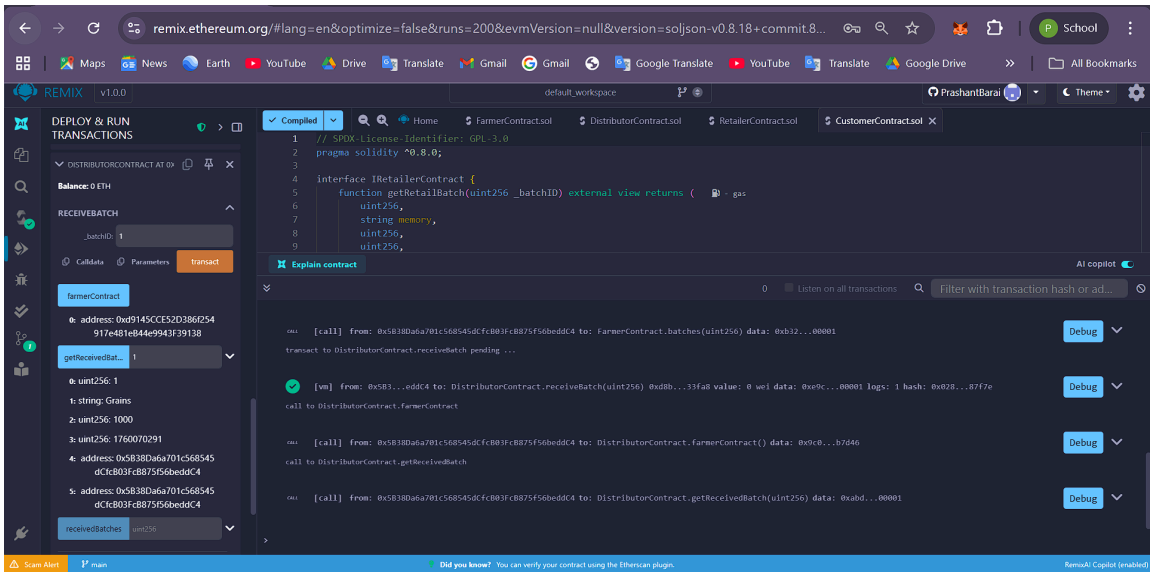


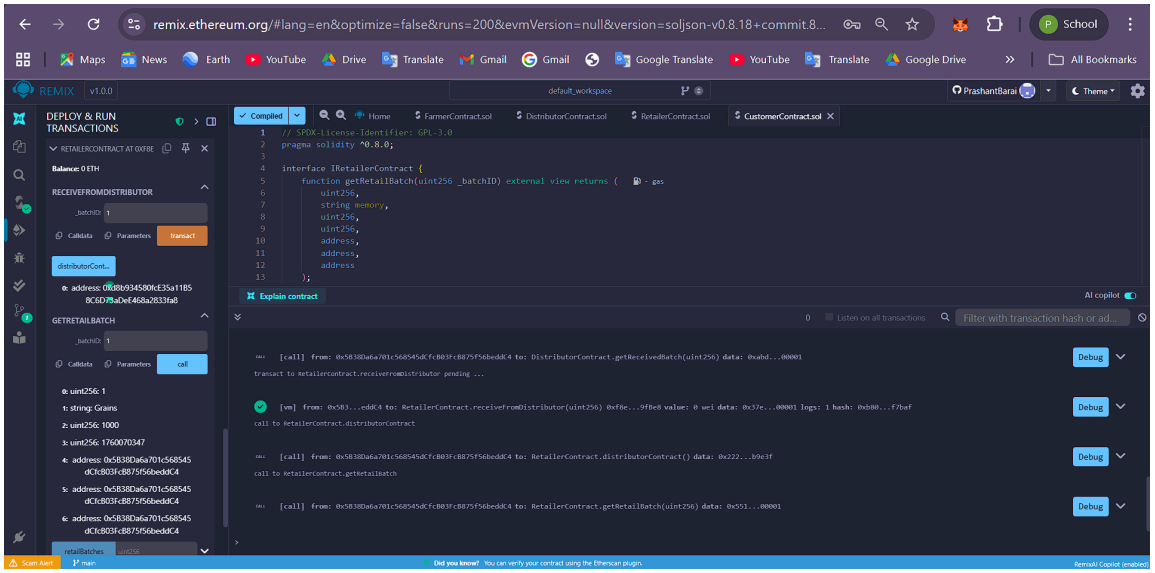


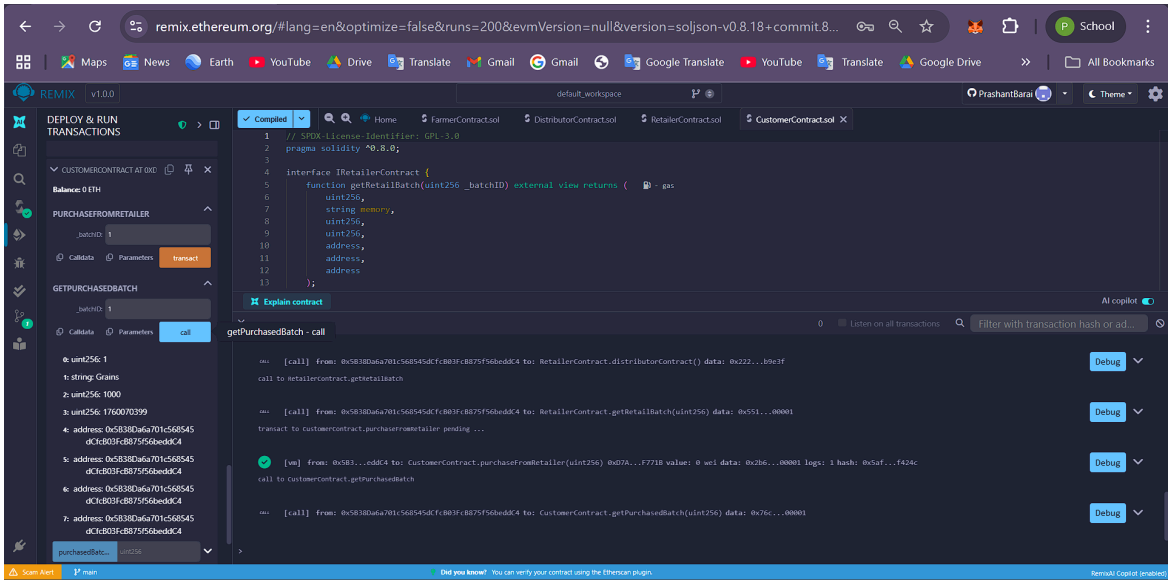


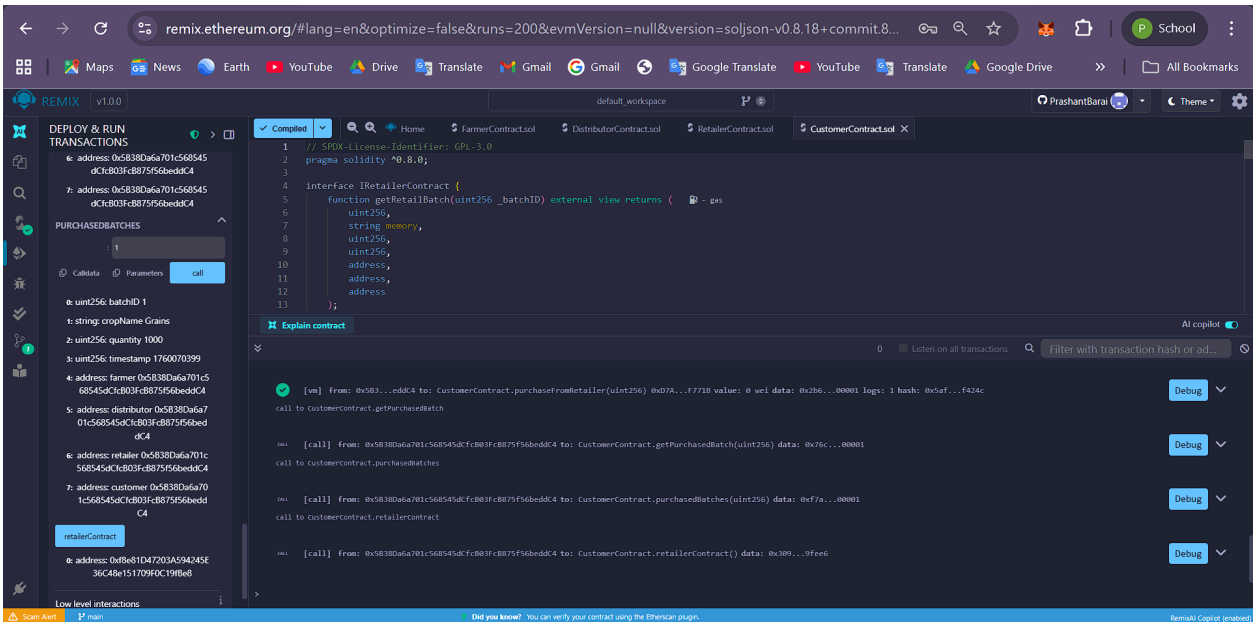


Prajyot Shinde *D20A* *Roll no.57*









Conclusion:

This experiment provided a clear understanding of how smart contracts automate processes and execute securely on the Ethereum blockchain. It demonstrated how predefined logic embedded within contracts can run autonomously without the need for intermediaries, ensuring transparency, reliability, and tamper-resistant execution. By observing how transactions trigger contract functions and how results are recorded immutably on the blockchain, the experiment reinforced key concepts such as decentralization, trustless systems, and data integrity.

Using Solidity along with Remix IDE made the entire process of development, deployment, and interaction both simple and efficient. Remix provides an intuitive interface for writing, compiling, and debugging contracts, while Solidity enables developers to define complex logic in a structured and secure manner. This combination allows even beginners to quickly grasp the lifecycle of a smart contract, from creation to execution. Furthermore, the experiment highlighted the importance of testing and iteration in building robust decentralized solutions. It showed how developers can refine contract logic, identify potential vulnerabilities, and ensure optimal performance before deploying to a live network. Overall, this hands-on experience forms a strong foundation for understanding and developing decentralized applications (dApps), equipping learners with the essential skills needed to explore advanced blockchain concepts and real-world implementations.